

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0306

Sierra Blanca 100 kV
July 30, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0306. The Interconnection Customer proposes a 116.0 MW natural gas combined cycle facility interconnecting at the Sierra Blanca 100 kV substation in the Illinois area. The request was received on February 04, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 188 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,113,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SIERRA BLANCA / CONRAD AUTO 100kV
Contingency: MINNEAPOLIS CITY OF / WALLENPAUPACK 500kV
Loading: 114.0% (Rating: 125 MVA)
DFAX: 100.0%
Pre-project loading: 21.3%
2. Facility: CLINTMOORE / CONRAD AUTO 100kV
Contingency: WALLENPAUPACK / MINNEAPOLIS CITY OF 500kV
Loading: 112.0% (Rating: 123 MVA)
DFAX: 100.0%
Pre-project loading: 17.8%
3. Facility: BEAR CANYON / CLINTMOORE 100kV
Contingency: CLINTMOORE / LAKE CHARLES PLANT 100kV
Loading: 107.2% (Rating: 121 MVA)
DFAX: 100.0%
Pre-project loading: 11.1%
4. Facility: ARCH SMITH / SWEATT 100kV Line
Contingency: ARCH SMITH / SWEATT 100kV Cir. 1
Loading: 100.3% (Rating: 189 MVA)
DFAX: 9.1%
Pre-project loading: 94.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SIERRA BLANCA / CONRAD AUTO 100kV	Rebuild line with higher-capacity conductor	\$4,734,000	100% (\$4,734,000)
CLINTMOORE / CONRAD AUTO 100kV	Rebuild line with higher-capacity conductor	\$4,830,000	100% (\$4,830,000)
BEAR CANYON / CLINTMOORE 100kV	Reconductor line	\$1,440,000	100% (\$1,440,000)
ARCH SMITH / SWEATT 100kV	Reconductor line	\$651,000	17% (\$109,000)
	Total allocated to project		\$11,113,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.